Fiscal Year:	FY 2012	**Task Last Updated:**	FY 06/13/2012
PI Name:	Wu, Honglu Ph.D.		
Project Title:	Dependence of radiation quality on charged particle-induced early and late damages in chromosomes		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI email changed from honglu.wu-1@nasa.gov after PI retirement from NASA.		
Project Type:	Ground	**Solicitation / Funding Source:**	2010 Space Radiobiology NNJ10ZSA001N
Start Date:	07/01/2011	**End Date:**	06/30/2014
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Zhang, Ye (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Tumor formation in humans or animals is a multi-step process. An early stage of cancer development is believed to be genomic instability (GI) which accelerates the mutation rate in the descendants of the cells surviving radiation exposure. A relationship between GI and tumor metastasis, as well as between GI and adaptation to drug treatment, has also been suggested. Evidence for GI induced from radiation exposure has been established in the last two decades. GI is defined as elevated or persistent genetic damages occurring many generations after the cells are exposed, and can be manifested in several biological consequences including mutation, apoptosis and damages in chromosomes. Genomic instability can be induced by both low- and high-LET radiation. Space radiation consists of a mixture of different types of ions covering a wide range of masses and energies. Its associated health risks to astronauts in flight are of special concern for prolonged space missions beyond the protective terrestrial magnetosphere. Before such missions can be undertaken, a much more detailed understanding of these biological effects is required to allow planning and implementation of

Task Description:	protective countermeasures. The proposed research will attempt to investigate GI induced from exposures to energetic ions of the types and energies encountered during spaceflight, and will address the critical question of carcinogenesis risk and its dependence on radiation quality which is an emphasized area of the current NASA Research Announcement (NRA). The overall goals of this proposed project are (1) to determine the relationship between the quality and fluence of energetic charged particles and the induction of delayed damages in chromosomes, which is commonly referred to as chromosomal instability and (2) to understand the relationship between chromosomal instability and other events at the cellular and molecular level that play a role in the maintenance of genome integrity. For this proposal, we select well-developed and widely used methodologies for quantification of GI. The three models include human T-lymphocytes collected from healthy subjects, and haemopoietic cells extracted from CBA/CaH and from C57BL/6J mice, which differ in radiosensitivity. The cells will be exposed in vitro to protons, Si and Fe ions which represent three of most important particle types in space. Chromosomal instability will be analyzed in both the total population of the cells surviving after multiple generations post irradiation (non-clonal assay), as well as in clonal populations derived from single cells surviving particle irradiation (clonal assay). The non-clonal assay is less labor intensive and allows analysis of samples exposed to particles of different types, energies and doses, whereas the clonal assay will be designed to tackle the question of genetic background influence on the induction of GI. Several techniques for chromosome analysis will be used which include multi-color fluorescence in situ hybridization (mFISH) and the multi-color banding in situ hybridization (mBAND). Measurements of telomere length and expressions of genes involved in genomic stability will also be performed in cells from unstable, as well as stable, clones. The clonal expansion assay is arguably more suitable than the non-clonal assay for determining the correlation between chromosomal instability and other biological consequences. Key to our proposal is that the investigators have substantial expertise and published literature to ensure the success of the proposed studies. The PI has extensive experience in the induction of heavy ion-induced chromosomal aberrations in human lymphocytes using mFISH, mBAND and other modern chromosome painting techniques. The Co-I was trained in immunology and has substantial expertise in T-lymphocyte isolation and expansion. Dr. Kadhim is a recognized expert in GI and had made significant contributions to the understanding of radiation-induced GI.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research addresses the question of genomic instability from charged particle exposures in space. By assessing genomic instability quantitatively for different charges and energies of the particles, we aim at contributing to the reduction of uncertainties in the assessment of cancer risks in astronauts after long-duration spaceflights.
Task Progress:	The following activities were undertaken: 1. Human lymphocytes were exposed to gamma rays and chromosome aberrations were analyzed in cells collected at different time points of cell culture. 2. Human lymphocytes and human lung epithelial cells were exposed to 600 MeV/u Fe ions at NASA Space Radiation Laboratory at Brookhaven National Laboratory for analysis of chromosome damages at different time points of cell culture. 3. Mouse bone marrow cells were exposed to 600 MeV/u Fe ions at NASA Space Radiation Laboratory at Brookhaven National Laboratory for analysis of chromosome damages at different time points of cell culture.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Abstracts for Journals and Proceedings	Sunagawa M, Mangala L, Zhang Y, Wilson B, Cucinotta F, Wu H. "Early and late damages in the chromosome of human lymphocytes after radiation exposure." Space radiation risks. Presented at 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, February 14-16, 2012. , Feb-2012